Lecture 5 – Lab 5

Space complexity

Algorithm Analysis: Time vs. Space Complexity

```
int x = 5;
int y = 7;
int z = x + y;
print(z);
if (z == 13) print("Done");
```

Focuses on executable lines/operations.

6 operations executed $\longrightarrow$ $T(n) = 6$

Result: O(1) (Constant Time)

Focuses **ONLY** on variable declarations.

$\rightarrow$ $S(n) = 3$

Result: O(1) (Constant Space)

Time counts how many times lines execute. **Space** counts how many memory containers are created.

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Find Space complexity?

```
int n = user_input;
for (int i = 0; i < n; i++) {
  // do something
}
```

```
int m = user_input;
int[] n = new int[m];
for (int i = 1; i <= m; i++) { print(n[i]); }
```

```
int Add(int n) {
  if (n <= 0) return 0;
  // Base case

  int variable = n;
  // 1 variable per
  level

  return n + Add(n - 1);
  // Recursive call
}
```

```
def pair_sum(a, b):
    x = a * 2   # Nested scope variable
    y = b + 1   # Nested scope variable
    return x + y

def add_sequence(n):
    sum = 0             # Base scope variable
    for i in range(n): # Base scope variable
        sum += pair_sum(i, i + 1)
    return sum

# Example call
add_sequence(10)
```

```
void exampleFunction(int n) {
    if (n <= 0) return;   // Base Case
    int x = 10;           // Variable 1
    int y = 20;           // Variable 2
    // n is also a variable passed in
    exampleFunction(n-1); // Recursive Call
}
```

```
int fact(int n){
  if (n == 1)
  {
     for(i=1, i<=n; i++)
     {
        print(i);
     }
    return 1;
  }
  else
    return fact( n-1 ) + fact( n-2 );
}
```

function prestiferous(n)

$r := 0$

for $i := 1$ *to* n *do*

for $j := 1$ *to* i *do*

for $k := j$ *to* $i + j$ *do*

for $l := 1$ *to* $i + j - k$ *do*

$r := r + 1$

return(r)

```
int F(A,n)
S=0;
for i = 0 to n-2
   for j = i+1 to n-1
         if (Ai > Aj) S++
return S
```

```
int fact(int n){
  if (n == 0)
    return 1;
  else
    return fact(n - 1);
}
```

```
int fact(int n){
  if (n == 1)
    return 1;
  else
    return fact( n/2 );
}
```